ParkScreen — Screening Report

Multimodal decision-aid for Parkinson's disease signs — *not a diagnosis*.

C

OVERALL RISK GRADE

Elevated likelihood of features consistent with PD

Fused score 0.70

Modality scores

Vocal

weight 87%

Speech channels: phonation 0.61, DDK 0.88.

Facial

weight 13%

Smile classifier 0.15. AU12 peak 2.61, blink 21.3/min.

Vocal channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Phonation 35% within modality	0.61	1.00	jitter 0.35%, HNR 25.7 dB, F0 SD 5.6 Hz
Articulation (DDK) 65% within modality	0.88	1.00	rate 3.85 syl/s, ISI CV 0.66, amp CV 0.40

Facial channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Smile task 100% within modality	0.15	1.00	AU12 peak 2.61, blink 21.3/min

Consistency

> **Channels partially agree**
> Speech channels align; facial signal is weaker but directionally consistent.

Clinical narrative

Risk Level

Elevated. The AUC-weighted fused PD probability is **0.704**, driven primarily by the speech channels. As screening evidence this places the recording above the threshold typically warranting further evaluation, but it is not a diagnosis.

Phonation

Voice-quality features are largely within typical adult ranges: jitter of **0.354%** and shimmer of **2.670%** sit at or below normative bounds, and HNR of **25.66 dB** reflects a clean harmonic signal with low noise. F0 mean of **162.4 Hz** with std **5.6 Hz** and range **35.0 Hz** suggests somewhat restricted pitch variability, which can be consistent with early prosodic flattening. The phonation channel probability of **0.613** reflects this borderline prosodic picture rather than overt perturbation.

Articulation (DDK)

The PATAKA task shows a markedly slowed diadochokinetic rate of **3.85 Hz**, well below the ≥6 Hz expected in healthy adults. Inter-syllable timing is highly irregular (**isi_cv 0.656**) and amplitude variability is elevated (**amp_cv 0.399**), with a mild negative amplitude decrement (–**0.0147**) suggesting slight fading across the trial. This combination of bradykinetic rate, dysrhythmia, and amplitude drift is consistent with PD-typical articulatory patterns and drives the high channel probability of **0.881**.

Facial Expression

Facial signals do not show hypomimia markers: smile amplitude on cue is preserved (**AU12 amplitude 2.61**, mean AU12 **1.55**), expression variance is healthy at **0.541**, and blink rate of **21.3/min** sits at the upper end of the normative 12–20/min baseline. Detection rate was **1.00** with no warnings, so the low channel probability of **0.148** appears to be a reliable read.

Cross-Channel Consistency

The two speech channels agree with each other, strengthening the articulatory/phonatory signal. However, the facial channel diverges substantially from the speech modalities — **per-channel disagreement, flagged for clinical review.** This pattern (elevated speech-based risk with preserved facial expressivity) is plausible in early or predominantly speech-affecting presentations, where bulbar/articulatory signs can emerge before overt hypomimia; it may also reflect task-specific compensation during the smile cue. The clinician should weigh both interpretations.

Recommended Next Steps

Given the elevated fused probability and the strong DDK findings, referral to a movement-disorder neurologist for in-person evaluation is recommended. Additional structured speech assessment (connected speech, reading passage) may help characterize the articulatory pattern further. The discordant facial channel should be reviewed rather than dismissed.

Disclaimers

This report is a screening decision-aid, not a clinical diagnosis. Any risk indication requires evaluation by a qualified movement-disorder specialist.

The classifiers were trained on Parkinson's-disease subjects recorded 2–5 hours post-medication (ON state). Off-state PD may present with more perturbed features than the training distribution, so this system's calibrated probability does not extrapolate to OFF-state inputs.